Disney Parks Industrial Engineering Case Study

Project overview: how the Autumn Harvest Festival models work and what they found

Case: Walt Disney Parks & Resorts, 2026 IE Case Study	**Team:** Aarnav Bhat, Eisen Hy, Justin Llamera	**Tools:** Excel

About this document. Our submission to Disney was a slide deck plus three Excel models. This overview was written afterward for my portfolio, to explain the methods and results in one place for anyone opening the models.

The problem

A regional fall festival runs four weekends a year (Saturdays and Sundays, 10 AM to 8 PM) with food, merchandise and family activity booths. 2026 is its 25th anniversary, with a bigger marketing budget, advertising in new cities and a new concert. Leadership asked three connected questions:

- **Part 1.** How many guests will come in 2026, and on which weekend will the crowd peak?
- **Part 2.** How should 120 booth spaces be split between food, merchandise and activities?
- **Part 3.** To grow, should the festival stay open longer each day, add Fridays, or add a fifth weekend?

Each answer feeds the next. The Part 1 forecast sets the busiest day, Part 2 sizes the booths for that day, and Part 3 relieves the bottleneck Part 2 uncovers.

68,000
2026 visits forecast (+21% vs. 2025)

10,808
guests on the expected peak Saturday

45 / 21 / 27
food, merchandise, activity booths

$55K
yearly net gain from 12-hour days

Part 1: Attendance forecast

> **Answer:** plan for about **68,000 visits** (likely range 66,300 to 70,300). Weekend 4 becomes the peak at about 20,800 visits because of the concert, with Weekend 3 close behind, so both should be staffed as peaks.

Method

- **A trend baseline from 5 simple models.** With only five years of history (2021–2025), complex methods like ARIMA or machine learning can't be fitted reliably. We averaged five simple trend methods (linear, log-linear, two growth-rate calculations and average yearly growth). They agreed within about 1,650 visits, giving a baseline of 57,800.
- **Backtesting.** Each method was tested by forecasting 2024 and 2025 from earlier years. Every method came in too low, which showed that history alone underestimates this growing festival.
- **2026 adjustments on top of the trend.** The marketing increase used a published advertising elasticity. Guests from the newly targeted cities were estimated from population times a reduced attendance rate for distance. The anniversary boost and concert-only guests were small labeled judgments. Together they add about 10,200 visits.

- **Uncertainty.** Low, expected and high scenarios, plus a 1,000-run Monte Carlo simulation that lets the drivers offset each other.

Scenario	Total visits	Growth vs. 2025	Peak weekend	Peak Saturday
Low	60,700	+8%	Weekend 3	9,200
Expected	**68,000**	**+21%**	**Weekend 4**	**10,800**
High	77,100	+38%	Weekend 4	13,500

Also flagged: the concert venue seats 8,000, so at two performances it fits roughly 77% of expected Weekend 4 guests. We recommended timed reservations or an overflow viewing area.

Part 2: Booth mix

> **Answer: 45 food, 21 merchandise and 27 activity booths.** Activity booths take double space, so this fills all 120 spaces. On the expected peak day, no line is longer than about 6 minutes, vendors take in about $394,000 in sales, and the mix earns more than any other that keeps lines short.

Method

- **Step 1: the minimum booths to keep up.** The peak day from Part 1 was turned into guests per hour for each booth type. A standard queueing model (each booth treated as its own line) gave the fewest booths needed to keep every booth at or below 85% busy: 41 food, 14 merchandise, 23 activity. That uses 101 of the 120 spaces.
- **Step 2: the best use of the rest.** We scored every possible mix (about 900) and kept only those with short lines and healthy variety. From those we picked the one earning the most after vendor costs.
- **Step 3: stress test.** The chosen mix works on normal and slow days. On a high-scenario Saturday of 13,500 guests, food booths run over 100% busy. That is a capacity problem, not a mix problem, and it sets up Part 3.

Mix (food / merch / activities)	Longest wait	Vendor sales per day	Keeps lines short?
Recommended: 45 / 21 / 27	**6 min**	**$394,000**	**Yes**
Match crowd shares: 52 / 22 / 23	12 min	$374,000	No
Equal space: 40 / 40 / 20	45 min	$353,000	No
Food-heavy: 60 / 36 / 12	Lines never clear	$338,000	No

Why not fill the grounds with food? Food earns the most per space, but too few activity booths means very long activity lines and shorter family visits. Activities keep families on site longer, and longer visits mean more food sales. The food-heavy mix actually earns less. Vendor cost percentages were tested up and down, and the recommended booth counts didn't change.

Part 3: Growth options

> **Answer: stay open two hours later (10 AM to 10 PM) starting in 2027.** It is the cheapest option, the only one that makes money whether crowds are small or large, and the only one that shortens Saturday lunch lines. Keep a fifth weekend in reserve for later. Don't add full Fridays.

Method

- **Count only truly new visitors.** A Friday or fifth-weekend visitor only counts if they wouldn't have come anyway. We assumed 40% of Friday visitors and 30% of fifth-weekend visitors had simply moved their visit.
- **Build costs from the ground up.** A festival day costs about $27,000 (staff, security, entertainment, facilities, permits, parking, marketing). Only some costs grow with longer hours, so two extra hours cost about $4,000 a day, not a fifth of $27,000.
- **Test the risk.** For each option: break-even visitors, five-year net present value under low, expected and high demand, and a booth-level profit check for vendors.

Expected case	A: 12-hour days	B: add Fridays	C: fifth weekend
New visitors per year	5,400	9,500	7,100
Added cost per year	$32,000	$97,000	$84,000
Net gain per year	**$55,000**	**$36,000**	**$3,000**
If crowds are smaller than expected	+$12,000	−$11,000	−$28,000
5-year NPV (expected)	$205,000	$143,000	$6,000
Peak food-booth use (from 82%)	76%	74%	78%

Why not Fridays? At a workday crowd (about 45% of a Saturday), a typical food booth takes in about $2,270 but spends about $2,310 on food and a full day of staff. Vendors who lose money on Fridays stop showing up, and half-empty booths would hurt the festival's brand.

Key assumptions and limits

- The case gives five years of history, no ticket price, no vendor fee structure and no booth service speeds. Each missing number is a single labeled input in the workbooks, so it can be changed and the results update.
- Assumed values include a $10 admission value per visitor, a 15% festival share of vendor sales (matching a comparable five-weekend fall festival), vendor cost rates of 65% (food), 55% (merchandise) and 40% (activities), and the visitor-shift percentages above.
- The biggest source of uncertainty is how many people in the newly targeted cities will come in year one.

Files in this project

File	What it contains
Final slide deck (PDF)	The 46-slide presentation we submitted
Case prompt (PDF)	Disney's original case questions and data
festival_forecast_model.xlsx	Part 1: trend models, backtest, assumptions, scenarios and Monte Carlo simulation
booth_mix_model.xlsx	Part 2: peak-hour demand, queueing capacity check and every booth mix scored
growth_options_model.xlsx	Part 3: new visitors, costs, break-even, 5-year NPV and the weighted decision score for each option